Aircraft Noise: Classifier Training Summary

DESE71003 Sensing & IoT coursework · first training run on one week of data (23–29 Oct 2026) · written 31 Oct 2026

1. Goal

v1 of the system announces every aircraft that was actually *heard*. ADS-B tells us which aircraft are nearby, but not whether they were audible indoors. The classifier is the "did the microphones hear an aircraft?" gate: it looks only at the sound levels from the two ESP32/INMP441 nodes and outputs a probability. A positive result is then matched to the closest ADS-B pass before an announcement is made.

2. Data and labelling (model/dataset.py)

Inputs, privacy-preserving by design. Each node sends one packet per second: LAeq, LAmax and 7 octave bands (63 Hz–4 kHz), and never any audio. One example is a **causal 30 s window** of 9 channels × 30 seconds, so it uses only the past and can run live. From each window the mic's background level is subtracted: the median of the 3 whole minutes before the window. Every value is therefore "dB above the usual level", which removes most of the day/night and room differences.

Labels come from ADS-B, never from the sound itself.

- **Aircraft (1):** windows ending between 5 s before and 15 s after the *sound arrival* (closest-approach time + slant distance / 343 m/s) of a pass within 3 km slant. Every 2nd second is kept.
- **Quiet (0):** no airborne aircraft within 8 km slant for ±45 s, and the ADS-B logger running at the time. Every 10th second is kept.
- **Left out:** the ambiguous zone (aircraft 3–8 km away), logger gaps, gaps of more than 2 s, mic-fault flags, and windows without a background. Single lost packets are filled with the last value.

Distances use the geoid-corrected GPS altitude and the receiver height. Ground traffic is filtered out, and helicopters (ADS-B category A7) are tracked separately.

Day	Aircraft windows	Distinct passes	Quiet windows	of which daytime (07–23)	Role
23 Oct	9,962	464	6,233	1,067	train
24 Oct	13,780	627	4,000	64	train
25 Oct	13,705	624	4,448	200	train
26 Oct	13,964	647	4,314	190	train
27 Oct	14,550	664	4,541	394	train
28 Oct	14,737	671	4,858	466	validation
29 Oct	6,050	275	6,870	2,612	test

122,012 windows in total, from 3,972 passes (counted per day). **86% of quiet windows are at night, but only 9% of aircraft windows are.** Heathrow is busy from 06:00 to 23:00, so a genuinely quiet daytime minute is rare. This imbalance is the main weakness of the dataset (see §5).

3. Models and training (model/train.py)

- **Split by whole days:** 5 days train / 1 day validation / 1 day test. The data is strongly autocorrelated (neighbouring seconds are near-copies), so a random split would leak test windows into training and inflate every score.
- **Threshold rule (baseline 1):** "LAmax rose ≥ T dB above the background in the last 10 s". This is the obvious non-ML detector.
- **Gradient boosting (baseline 2):** scikit-learn HistGradientBoosting on 36 summary features (per channel: peak and mean of the last 15 s, rise from the start of the window, spread), with class-balanced weights.
- **1D CNN:** 13,281 parameters. Each channel and its first difference (dB/s) go into three dilated convolutions with receptive fields of 5, 13 and 29 s. Mean and max pooling make it position-invariant. Trained with class-weighted BCE and AdamW (lr 1e-3), 15 epochs, keeping the epoch with the best validation loss. Training took about 10 minutes on the Mac CPU.
- **Operating threshold:** for each model, the threshold that maximises F1 on the validation day is chosen, then applied unchanged to the test day. Channels are scaled by their training-set spread.

4. Results (test day 29 Oct, never seen during training or tuning)

Detector	Precision	Recall	F1	False alarms (% of quiet s)	Passes detected (1–2 km)
Threshold rule (≥1.9 dB)	0.62	0.94	0.74	51.4%	98% of 484
Gradient boosting	0.85	0.98	0.91	14.9%	100% of 484
1D CNN	0.90	0.98	0.94	9.1%	100% of 484

- CNN ROC AUC is **0.991** over all test windows and **0.975** on daytime windows only. Both mics perform the same (precision 0.89 / 0.91, recall 0.98).

- Pass-level detection is close to 100% at every distance, and 98% for the 54 helicopter mic-pass pairs (each pass is counted once per mic).
- **False alarms are concentrated in the daytime:** 16–22% of quiet daytime seconds, against 1–4% at night.
- **Daytime-only training (07–23 h) did worse:** CNN F1 0.89, with false alarms at 42% of quiet seconds. Only 1,915 quiet daytime windows were available for training and 466 for validation, so the F1-chosen threshold collapsed (the threshold rule even picked −2.7 dB). This confirms that the bottleneck is the amount of quiet daytime data, not the model.

5. Is this the best way to train it? Assessment against Lecture 4

The overall approach is sound, and it follows the lecture's advice closely:

- **Start simple, compare with classical methods:** two baselines are reported. Boosting on hand-made features reaches F1 0.91 against the CNN's 0.94, an example of "classical methods often work surprisingly well".
- **Autocorrelation:** recognised and handled with whole-day splits. This is the most common mistake in time-series ML and it is avoided here.
- **Think in time and frequency:** the octave bands are an on-device spectral decomposition, and the CNN uses both levels and their rate of change.
- **Privacy-preserving analytics:** only band levels leave the node, so speech cannot be reconstructed from them.

What is not yet best practice, and what to change:

- **Day/night confound (most important).** Because quiet windows are mostly at night, a model could partly learn "busy-daytime texture = aircraft". Background subtraction reduces this, and the daytime AUC of 0.975 suggests the effect is limited. The fix is data: the second collection week adds daytime quiet windows. Also report daytime metrics as the headline number.
- **One split, one seed, so no error bars.** The lecture's points on sample size and variance (CLT; Vabalas et al. 2019 on validation with limited samples) argue for **leave-one-day-out cross-validation**: 7+ folds, each day tested once, reporting mean ± sd. The current test day is unusual: 29 Oct has 275 passes against about 650 on other days, and its aircraft/quiet balance is 47/53 against about 75/25. A single-day score can therefore mislead.
- **Effective sample size is passes, not windows.** The 66k aircraft windows come from 3,026 training passes, which is still well above the lecture's "~1,000 per class" rule of thumb. The 1,915 daytime quiet windows, however, are only a few hundred independent quiet periods, and that is below it.
- **Choose the threshold for a target false-alarm rate, not maximum F1.** F1 depends on the class balance, which differs between days. Pick, for example, ≤2% false alarms on validation, and report ROC and precision-recall curves rather than a single point.
- **The test set is cleaner than real use.** Aircraft 3–8 km away are excluded from evaluation, but the deployed system will see them. Add a **continuous-stream evaluation** on the test day: run the model every second, merge consecutive detections into events, and count false events per hour plus missed passes. This is the number the announcer actually experiences.
- **Split by runway mode.** Heathrow alternates runways (see the alternation PDF) and switches between west and east operation with the wind. Validation and test days should each include an easterly day, or the model will be evaluated on traffic patterns it never trained on.
- **Use both mics together (cross-correlation, Granger-style reasoning).** An aircraft raises both rooms at nearly the same time, while a door or TV usually affects one. A two-mic input (18 channels), or a rule requiring agreement, should cut daytime false alarms. Cross-correlating the mic level with the ADS-B distance curve also checks the 343 m/s arrival-time alignment that the labels rely on.
- **Document the data:** add a short "datasheet for the dataset" (Gebru et al., cited in the lecture) to the report appendix, covering collection, labelling rules, exclusions, known biases (night/day) and privacy.

6. Bottom line

One week is **enough for the aircraft class** but **not enough quiet daytime data**. The model choice (a small CNN with two baselines) and the day-wise split are appropriate, so the gains now come from evaluation and data rather than a bigger network. Next steps, in priority order: (1) collect week 2; (2) leave-one-day-out CV with error bars and daytime-only metrics; (3) a false-alarm-targeted threshold and a continuous-stream event-level test; (4) try two-mic fusion. In deployment, ADS-B fusion removes most remaining false alarms, because a detection with no aircraft nearby is never announced.